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Collisionless Damping of Nonlinear Plasma Oscillations

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It is well known that the linear theory of collisionless damping breaks down after a time $\tau \equiv (m/e\epsilon k)^{1/2}$, where k is the wavenumber and ϵ is the amplitude of the electric field. Jacobi elliptic functions are now used to provide an exact solution of the Vlasov equation for the resonant electrons, and the damping coefficient is generalized to be valid for times greater than $t = \tau$. This generalized damping coefficient reduces to Landau's result when $t/\tau \ll 1$; it has an oscillatory behavior when t/τ is of order unity, and it phase mixes to zero as t/τ approaches infinity. The above results are all shown to have simple physical interpretations.

I. REVIEW OF THE LINEAR THEORY OF COLLISIONLESS DAMPING

THE basic equations for the problem are the Vlasov equation for the electron distribution and Poisson's equation. Dawson's model of collisionless damping¹ also uses the equation expressing conservation of energy which is easily derived from the above two equations. The ions cannot participate in the high-frequency plasma oscillations and just form a uniform background charge. Also, the coordinates perpendicular to the propagation vector of the wave may be integrated out of the Vlasov equation at the outset; so, it is only necessary to consider the problem in one dimension.

To explain the mechanism of collisionless damping, Dawson divides the electron distribution into a main part and a resonant part (see Fig. 1). He shows that the main part of the distribution supports the oscillatory motion of the plasma wave and that the resonant part of the distribution damps the wave. To get the damping coefficient, he first calculates the rate of increase of the kinetic energy of the resonant electrons. By invoking the conservation of energy, he sets this rate of increase of kinetic energy equal to the rate of decrease of wave energy. The latter quantity immediately gives the damping coefficient of the wave.

To make these ideas more quantitative, consider a plasma wave with electric field²

$$\epsilon(x, t) = \epsilon \sin(kx - \omega t). \quad (1)$$

If we work in a coordinate system moving with the wave, then the electric field can be written as

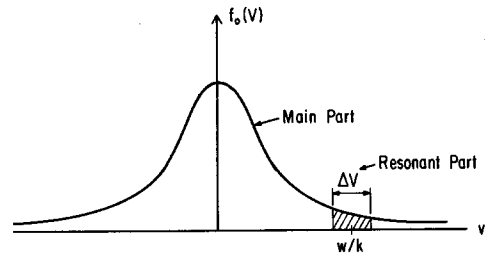


FIG. 1. The division of the electron distribution into a main part and a resonant part.

$\epsilon \sin(kx)$ and the rate of increase of kinetic energy density can be written as

$$\frac{dT}{dt} = \frac{nm}{2} \int_{-\lambda/2}^{\lambda/2} \frac{dx}{\lambda} \int_{-\infty}^{+\infty} dv \left(v + \frac{\omega}{k} \right)^2 \frac{\partial f}{\partial t}, \quad (2)$$

where nm is the electron density, λ is the wavelength, and f is the electron distribution in the waveframe. The x integration just gives the spatial average of f ; so, Eq. (2) becomes

$$\frac{dT}{dt} = \frac{nm}{2} \int_{-\infty}^{+\infty} dv \left(v + \frac{\omega}{k} \right)^2 \frac{\partial \langle f \rangle}{\partial t}. \quad (3)$$

The time derivative of $\langle f \rangle$ is easily obtained from the spatial average of the Vlasov equation

$$\frac{\partial \langle f \rangle}{\partial t} = \left\langle \frac{e}{m} \epsilon \sin(kx) \frac{\partial f}{\partial v} \right\rangle. \quad (4)$$

Let $f = f_0 + f_1$, where f_0 is the unperturbed part of the distribution and f_1 is the perturbation caused by the wave. Since f_0 is spatially homogeneous, it makes no contribution to Eq. (4). To get f_1 , we may use the linearized Vlasov equation

$$(\partial f_1 / \partial t) + v (\partial f_1 / \partial x) = (e/m) \epsilon \sin(kx) (\partial f_0 / \partial v). \quad (5)$$

The solution of Eq. (5) is

$$f_1(x, v, t) = f_1(v, 0) \cos(kx - kv t) - (e/m) \epsilon (\partial f_0 / \partial v) \cdot \{ [\cos(kx)/kv] - [\cos(kx - kv t)/kv] \}, \quad (6)$$

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¹ J. Dawson, *Phys. Fluids* **4**, 869 (1961).

² The formalism used in this section is different from that used by Dawson; however, it is quite similar to the quasi-linear formalism of W. E. Drummond and D. Pines.

where $f_1(v, 0)$ is the velocity dependence of the initial perturbation. Substituting Eq. (6) into the right-hand side of Eq. (4) gives

$$\frac{\partial \langle f \rangle}{\partial t} = \frac{\partial}{\partial v} \left\{ \frac{1}{2} \frac{e}{m} f_1(v, 0) \varepsilon \sin(kvt) + \frac{1}{2} \left(\frac{e}{m} \right)^2 \frac{\partial f_0}{\partial v} \varepsilon^2 \frac{\sin(kvt)}{kv} \right\}. \quad (7)$$

Substituting Eq. (7) into Eq. (3) and integrating by parts gives

$$\frac{dT}{dt} = -\frac{ne\varepsilon}{2} \int_{-\infty}^{+\infty} dv f_1(v, 0) \left(v + \frac{\omega}{k} \right) \sin(kvt) - \frac{ne^2\varepsilon^2}{2m} \int_{-\infty}^{+\infty} dv \frac{\partial f_0}{\partial v} \left(v + \frac{\omega}{k} \right) \frac{\sin(kvt)}{kv}. \quad (8)$$

After a few cycles (i.e., $kvt \simeq \omega t \gg 1$), the first integral phase mixes to zero and the integrand of the second integral becomes very peaked about $v = 0$. The width of this peaked function is of order $\Delta v \simeq 1/kt = (\omega/k)(1/\omega t)$. Only those electrons within Δv of $v = 0$ make any significant contribution to the rate of change of kinetic energy; so, Δv is the natural width of the resonant part of the distribution. Replacing the peaked function $\sin(kvt)/kv$ by $\pi\delta(kv)$ gives

$$\frac{dT}{dt} = -\frac{\pi}{2} \omega \frac{\omega_p^2}{k^2} \frac{\partial f_0}{\partial v} \bigg|_{v=0} \frac{\varepsilon^2}{4\pi}. \quad (9)$$

By equating the above rate of increase of kinetic energy to the rate of decrease of wave energy, $-(\partial/\partial t)(\varepsilon^2/8\pi) = -\gamma\varepsilon^2/4\pi$ and by realizing that $\partial f_0/\partial v|_{v=0}$ is equal to the derivative of the lab-frame distribution $\partial f_0'/\partial v|_{v=-\omega/k}$, we find exactly Landau's formula for the damping coefficient

$$\gamma_L = \frac{\pi}{2} \omega \frac{\omega_p^2}{k^2} \frac{\partial f_0'}{\partial v} \bigg|_{v=-\omega/k}. \quad (10)$$

This formula ceases to be valid when the linearized solution to the Vlasov equation breaks down (i.e., when $\partial f_1/\partial v$ becomes the same order of magnitude as $\partial f_0/\partial v$). Taking the derivative of Eq. (6) with respect to v produces secular terms growing like t in the main part of the distribution. The denominator of the second term in Eq. (6) goes to zero at $v = 0$; so, we must expand the numerator of this term about $v = 0$ before we can make sense out of the derivative in this region. Performing this expansion gives the following expression for f_1 in the resonant region

$$f_1(x, v, t) = f_1(v, 0) \cos(kx - kvt) + (e/m)t(\partial f_0/\partial v) \varepsilon \sin(kx) - (e/m)(kvt^2/2)(\partial f_0/\partial v) \varepsilon \cos(kx) + \dots \quad (11)$$

Taking the derivative of Eq. (11) with respect to v produces a secular term growing like t^2 . After a few cycles this term dominates, and it eventually causes the breakdown of the linear solution. If we call $t = \tau$ the time at which this breakdown occurs, then we find $\partial f_0/\partial v \sim \partial f_1/\partial v \sim (e/m)\varepsilon k\tau^2(\partial f_0/\partial v)$, or $\tau = (m/e\varepsilon k)^{1/2}$. Since the secular terms in the main part of the distribution grow like t rather than t^2 , the linear solution does not break down there until $t \simeq \tau(\omega\tau) \gg \tau$.

II. NONLINEAR THEORY OF COLLISIONLESS DAMPING

We can extend the theory of collisionless damping to times greater than $t = \tau$ by solving the exact Vlasov equation in the resonant region. To see that this is possible, we first note that the exact Vlasov equation just expresses incompressible flow in phase space. In the waveframe, its solution can be written as

$$f(x, v, t) = f[x_0(x, v, t), v_0(x, v, t), 0], \quad (12)$$

where $f[x_0, v_0, 0]$ is the initial distribution and (x_0, v_0) is the point from which (x, v) evolved. The evolution of this point is governed by the equation

$$m\ddot{x} = -e\varepsilon \sin(kx), \quad (13)$$

with initial conditions $(x, v) = (x_0, v_0)$. This equation is exactly soluble in terms of elliptic functions. The first integral of the equation is just the expression for conservation of energy

$$(m\dot{x}^2/2) - (e\varepsilon/k) \cos(kx) = W. \quad (14)$$

If we use the transformations $kx \equiv 2\xi$ and $\kappa^2 \equiv 2e\varepsilon/(kW + e\varepsilon)$, then Eq. (14) becomes

$$\dot{\xi}^2 = (1/\kappa^2\tau^2)[1 - \kappa^2 \sin^2(\xi)]. \quad (15)$$

When $\kappa^2 < 1$, Eq. (15) may be integrated to give

$$F(\kappa, \xi_0) = F(\kappa, \xi) - (t/\kappa\tau), \quad (16)$$

where $F(\kappa, \xi)$ is the elliptic integral of the first kind and κ takes the sign of ξ . When $\kappa^2 > 1$, it is useful to transform Eq. (15) in the following way:

$$\kappa \sin(\xi) = \sin(\zeta), \quad \dot{\xi}^2 = (1/\tau^2)[1 - (1/\kappa^2) \sin^2(\zeta)]. \quad (17)$$

Equation (17) may now be integrated to give

$$F(1/\kappa, \zeta_0) = F(1/\kappa, \zeta) - (t/\tau). \quad (18)$$

From Eq. (6) it follows that the initial distribution can be written as

$$f[x_0, v_0, 0] = f_0(v_0) + f_1(v_0, 0) \cos(kx_0). \quad (19)$$

To calculate the rate of increase of kinetic energy, we will be interested in

$$\begin{aligned} \frac{\partial f}{\partial t} &= \frac{\partial f}{\partial x_0}(x_0, v_0, 0) \frac{\partial x_0}{\partial t} + \frac{\partial f}{\partial v_0}(x_0, v_0, 0) \frac{\partial v_0}{\partial t} \\ &= f_1(v_0, 0)kv_0 \sin(kx_0) \\ &\quad + \left[\frac{\partial f_0}{\partial v_0} + \frac{\partial f_1}{\partial v_0} \cos(kx_0) \right] \frac{e}{m} \varepsilon \sin(kx_0). \end{aligned} \quad (20)$$

From the linear theory, we know that the contribution from $f_1(v, 0)$ phase mixes to zero in a time of order $(1/\omega)$. However, the linear theory breaks down in the resonant region, and we must give an auxiliary estimate of the contribution from $f_1(v, 0)$ in this region. From Poisson's equation, it follows that $k\varepsilon \sim ne \int dv f_1(v, 0) \sim nef_1(v, 0)v_{th}$ or $f_1(v, 0) \sim (1/\omega\tau)^2 1/v_{th} \sim (1/\omega\tau)^2 f_0$. Consequently, we find that

$$(\partial f_1/\partial v_0) \sim (1/\omega\tau)^2 (\partial f_0/\partial v_0),$$

$$f_1(v, 0)kv_0 \sin(kx_0) \sim (1/\omega\tau)(e/m)(\partial f_0/\partial v_0) \varepsilon \sin(kx_0).$$

Thus, we can write Eq. (20) as

$$\partial f/\partial t = (\partial f_0/\partial v_0)(e/m) \varepsilon \sin(kx_0)[1 + O(1/\omega\tau)].$$

If we treat $\partial f_0/\partial v_0$ as if it is constant in the above

equation, then the error incurred is also of order $(\partial^2 f_0/\partial v^2)\Delta v \sim (\partial f_0/\partial v)(\Delta v/v_{th}) \sim (1/\omega\tau)(\partial f_0/\partial v)$. By neglecting terms down by an order of $(1/\tau\omega)$, we can write the above equation as

$$\partial f/\partial t = (\partial f_0/\partial v_0)(e/m) \varepsilon \sin(kx_0), \quad (21)$$

where $\partial f_0/\partial v_0$ is evaluated at $v_0 = 0$. When $\kappa^2 < 1$, $\sin kx_0$ can be expressed as

$$\begin{aligned} \sin(kx_0) &= 2 \sin(\xi_0) \cos(\xi_0) \\ &= 2 \operatorname{sn}[F(\kappa, \xi_0), \kappa] \operatorname{cn}[F(\kappa, \xi_0), \kappa] \\ &= 2 \operatorname{sn}[F(\kappa, \xi) - (t/\kappa\tau), \kappa] \operatorname{cn}[F(\kappa, \xi) - (t/\kappa\tau), \kappa]. \end{aligned} \quad (22)$$

When $\kappa^2 > 1$, $\sin kx_0$ can be expressed as

$$\begin{aligned} \sin(kx_0) &= 2 \sin(\xi_0)[1 - \sin^2(\xi_0)]^{1/2} \\ &= 2/\kappa \sin(\xi_0)[1 - 1/\kappa^2 \sin^2(\xi_0)]^{1/2} \\ &= 2/\kappa \operatorname{sn}[F(1/\kappa, \xi_0), 1/\kappa] \operatorname{dn}[F(1/\kappa, \xi_0), 1/\kappa] \\ &= 2/\kappa \operatorname{sn}[F(1/\kappa, \xi) - (t/\tau), 1/\kappa] \\ &\quad \cdot \operatorname{dn}[F(1/\kappa, \xi) - (t/\tau), 1/\kappa]. \end{aligned} \quad (23)$$

By using the double angle formula for the Jacobi elliptic functions, we find when $\kappa^2 < 1$

$$\begin{aligned} \frac{\partial f}{\partial t} &= 2 \frac{e}{m} \varepsilon \frac{\partial f_0}{\partial v} \left\{ \frac{\sin(\xi) \operatorname{cn}(t/\kappa\tau, \kappa) \operatorname{dn}(t/\kappa\tau, \kappa) - \cos(\xi) \operatorname{sn}(t/\kappa\tau, \kappa)[1 - \kappa^2 \sin^2(\xi)]^{1/2}}{1 - \kappa^2 \operatorname{sn}^2(t/\kappa\tau, \kappa) \sin^2(\xi)} \right\} \\ &\quad \cdot \left\{ \frac{\cos(\xi) \operatorname{cn}(t/\kappa\tau, \kappa) + \sin(\xi) \operatorname{sn}(t/\kappa\tau, \kappa) \operatorname{dn}(t/\kappa\tau, \kappa)[1 - \kappa^2 \sin^2(\xi)]^{1/2}}{1 - \kappa^2 \operatorname{sn}^2(t/\kappa\tau, \kappa) \sin^2(\xi)} \right\}, \end{aligned} \quad (24)$$

where $\kappa^2 = [\xi^2 \tau^2 + \sin^2(\xi)]^{-1}$ and κ takes the sign of ξ . By using the definition in Eq. (17) and the double angle formula for the Jacobi elliptic functions, we find when $\kappa^2 > 1$

$$\begin{aligned} \frac{\partial f}{\partial t} &= 2 \frac{e}{m} \varepsilon \frac{\partial f_0}{\partial v} \left\{ \frac{\sin(\xi) \operatorname{cn}(t/\tau, 1/\kappa) \operatorname{dn}(t/\tau, 1/\kappa) - \tau \xi \cos(\xi) \operatorname{sn}(t/\tau, 1/\kappa)}{1 - \operatorname{sn}^2(t/\tau, 1/\kappa) \sin^2(\xi)} \right\} \\ &\quad \cdot \left\{ \frac{\cos(\xi) \operatorname{dn}(t/\tau, 1/\kappa) + \xi \tau \operatorname{sn}(t/\tau, 1/\kappa) \operatorname{cn}(t/\tau, 1/\kappa) \sin(\xi)}{1 - \operatorname{sn}^2(t/\tau, 1/\kappa) \sin^2(\xi)} \right\}. \end{aligned} \quad (25)$$

The rate of change of kinetic energy is given by

$$\begin{aligned} \frac{dT}{dt} &= \frac{nm}{2} \int_{-\frac{1}{2}\lambda}^{\frac{1}{2}\lambda} \frac{dx}{\lambda} \int_{-\infty}^{+\infty} dv \left(v + \frac{\omega}{k} \right)^2 \frac{\partial f}{\partial t} \\ &= \frac{4nm}{k^3} \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{d\xi}{\pi} \int_{-\infty}^{+\infty} d\xi \left(\xi + \frac{\omega}{2} \right)^2 \frac{\partial f}{\partial t}. \end{aligned} \quad (26)$$

Those terms in $\partial f/\partial t$ which are odd functions of ξ make no contribution to the integral in Eq. (26). If we delete these terms from $\partial f/\partial t$ and realize that the remaining terms are odd functions of ξ , then Eq. (26) can be written as

$$\frac{dT}{dt} = \frac{16nm\omega}{\pi k^3} \int_0^\infty d\xi \int_0^{\frac{1}{2}\pi} d\xi \xi \left(\frac{\partial f}{\partial t} \right)', \quad (27)$$

where $(\partial f/\partial t)'$ is the expression obtained from $(\partial f/\partial t)$ by deleting the terms which are odd functions of ξ . By changing coordinates from (ξ, ξ) to (ξ, κ) , we can write Eq. (27) as

$$\begin{aligned} \frac{dT}{dt} &= \frac{16nm\omega}{\pi k^3 \tau^2} \left\{ \int_0^1 \frac{d\kappa}{\kappa^3} \int_0^{\frac{1}{2}\pi} d\xi \left(\frac{\partial f}{\partial t} \right)' \right. \\ &\quad \left. + \int_1^\infty \frac{d\kappa}{\kappa^3} \int_0^{\sin^{-1}(1/\kappa)} d\xi \left(\frac{\partial f}{\partial t} \right)' \right\}. \end{aligned} \quad (28)$$

By inserting the expression for $(\partial f/\partial t)'$ into these two integrals and by using the definition in Eq. (17), we can write Eq. (28) as

$$\frac{dT}{dt} = \gamma_L \frac{\varepsilon^2}{4\pi} \left\{ \frac{64}{\pi} \int_0^1 \frac{dk}{\kappa^3} \int_0^{\frac{1}{2}\pi} \frac{d\xi}{\pi} \operatorname{sn}(t/\kappa\tau, \kappa) \operatorname{cn}(t/\kappa\tau, \kappa) [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} \frac{[2 \sin^2(\xi) - 1 - \kappa^2 \operatorname{sn}^2(t/\kappa\tau, \kappa) \sin^2(\xi)]}{[1 - \kappa^2 \operatorname{sn}^2(t/\kappa\tau, \kappa) \sin^2(\xi)]^2} \right. \\ \left. + \frac{64}{\pi} \int_1^\infty \frac{dk}{\kappa^5} \int_0^{\pi/2} \frac{d\xi}{\pi} \frac{\operatorname{sn}(t/\tau, 1/\kappa) \operatorname{dn}(t/\tau, 1/\kappa) \cos^2(\xi) [(2/\kappa^2) \sin^2(\xi) - 1 - (1/\kappa^2) \operatorname{sn}^2(t/\tau, 1/\kappa) \sin^2(\xi)]}{[1 - (1/\kappa^2) \sin^2(\xi)]^{\frac{1}{2}} [1 - (1/\kappa^2) \operatorname{sn}^2(t/\tau, 1/\kappa) \sin^2(\xi)]^2} \right\}, \quad (29)$$

where γ_L is Landau's damping coefficient given in Eq. (10).

In Appendix A, the integrands of Eq. (29) are represented as Fourier series and the ξ and ζ integrations are carried out. The result is

$$\frac{dT}{dt} = -\gamma_L \frac{\varepsilon^2}{4\pi} \sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 dk \left\{ \frac{2n\pi^2 \sin\left[\frac{\pi n t}{\kappa F \tau}\right]}{\kappa^5 F^2 (1+q^{2n})(1+q^{-2n})} + \frac{(2n+1)\pi^2 \kappa \sin\left[\frac{(2n+1)\pi t}{2F\tau}\right]}{F^2 (1+q^{2n+1})(1+q^{-2n-1})} \right\}, \quad (30)$$

where

$$q \equiv \exp(\pi F'/F), \quad F \equiv F(\kappa, \frac{1}{2}\pi),$$

and

$$F' \equiv F[(1 - \kappa^2)^{\frac{1}{2}}, \frac{1}{2}\pi].$$

If we write ε^2 as $\varepsilon_0^2 \exp[2 \int_0^t \gamma(t') dt']$ and invoke conservation of energy, we find

$$\gamma(t) = \gamma_L \sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 dk \left\{ \frac{2n\pi^2 \sin\left[\frac{\pi n t}{\kappa F \tau}\right]}{\kappa^5 F^2 (1+q^{2n})(1+q^{-2n})} + \frac{(2n+1)\pi^2 \kappa \sin\left[\frac{(2n+1)\pi t}{2F\tau}\right]}{F^2 (1+q^{2n+1})(1+q^{-2n-1})} \right\}, \quad (31)$$

$$\int_0^t \gamma(t') dt' = \gamma_L \tau \sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 dk \left\{ \frac{2\pi(1 - \cos\left[\frac{\pi n t}{\kappa F \tau}\right])}{\kappa^4 F (1+q^{2n})(1+q^{-2n})} + \frac{2\pi\kappa(1 - \cos\left[\frac{(2n+1)\pi t}{2F\tau}\right])}{F(1+q^{2n+1})(1+q^{-2n-1})} \right\}. \quad (32)$$

To check Eqs. (31) and (32), consider the limiting case when $t/\tau \ll 1$. In this case, one can use the small κ values $q \simeq (4/\kappa)^2$ and $F \simeq \frac{1}{2}\pi$ to show that the largest term on the right-hand side of Eq. (31) is the $n = 1$ contribution to the first term. Thus, Eq. (31) becomes

$$\gamma(t) \simeq \gamma_L \left\{ \frac{64}{\pi} \int_0^1 dk \frac{2\pi^2 \sin\left[\frac{\pi t}{\kappa F \tau}\right]}{\kappa^5 F^2 (1+q^2)(1+q^{-2})} + O\left(\frac{t}{\tau}\right) \right\}. \quad (33)$$

The integral in Eq. (33) is peaked in a region of small κ and may be written as

$$\gamma(t) \simeq \gamma_L \left\{ \frac{2}{\pi} \int_0^1 \frac{dk}{\kappa} \sin\left(\frac{2t}{\kappa\tau}\right) + O\left(\frac{t}{\tau}\right) \right\}.$$

This immediately reduces to the linear result

$$\gamma(t) \simeq \gamma_L \{1 + O(t/\tau)\}.$$

By taking into account the κ dependence of F and κF , one can show that the integrals over

$$\sin[(2n+1)\pi t/2F\tau], \quad \sin[\pi n t/\kappa F\tau], \\ \cos[(2n+1)\pi t/2F\tau], \quad \text{and} \quad \cos[\pi n t/\kappa F\tau],$$

phase mix to zero as t/τ approaches infinity. Hence, Eqs. (31) and (32) become

$$\gamma(t = \infty) = 0, \quad (34)$$

$$\int_0^\infty \gamma(t) dt = \gamma_L \tau \sum_{n=1}^{\infty} \frac{64}{\pi} \int_0^1 dk \left\{ \frac{2\pi}{\kappa^4 F (1+q^{2n})(1+q^{-2n})} + \frac{2\pi\kappa}{F(1+q^{2n-1})(1+q^{1-2n})} \right\}.$$

In Appendix B, the series in Eq. (34) are summed and $\int_0^\infty \gamma(t) dt$ is shown to have the value

$$\int_0^\infty \gamma(t) dt = \gamma_L \tau \frac{64}{\pi} \int_0^1 dk \left\{ \frac{1}{\kappa^4} \left[\frac{E}{\pi} - \frac{\pi}{4F} \right] + \frac{\kappa}{\pi} [E + (\kappa^2 - 1)F] \right\} \\ = O(\gamma_L \tau),$$

where $E \equiv E(\kappa, \frac{1}{2}\pi)$ is the complete elliptic integral of the second kind.

A limitation on the above results derives from the fact that in both the linear theory and the nonlinear theory we solved the Vlasov equation by treating the amplitude of the electric field as if it is constant. However, as the wave damps or grows, the amplitude varies as $\varepsilon_0^2 \exp[2 \int_0^t \gamma(t') dt']$. Consequently, the present treatment is valid only if the nonlinear effects prevent the amplitude from changing by a significant amount. Since $\int_0^\infty \gamma(t) dt \simeq O(\gamma_L \tau)$, it follows that the above condition is satisfied when $\gamma_L \tau \ll 1$. The present treatment can be looked upon as an expansion in which the Vlasov equation and Poisson's equation are solved simultaneously to first order in the expansion parameter $\gamma_L \tau$. We have also retained only lowest-order terms in the small parameter $(1/\omega\tau)$.

The results expressed in Eqs. (31) and (32) are similar, but not identical, to the results of Altshul and Karpman's recent treatment of the present problem.³ These authors also treat the amplitude of the electric field in the Vlasov equation as if it is constant, but, even under this restriction, their solution of the Vlasov equation is only approximate. They first Fourier transform the spatial dependence of the Vlasov equation

$$\frac{\partial f_0}{\partial t} = \frac{e}{m} \varepsilon_k \frac{\partial f_{-k}}{\partial v} + \frac{e}{m} \varepsilon_{-k} \frac{\partial f_k}{\partial v},$$

$$\frac{\partial f_k}{\partial t} + i v k f_k = \frac{e}{m} \varepsilon_k \frac{\partial f_0}{\partial v} + \frac{e}{m} \varepsilon_{-k} \frac{\partial f_{2k}}{\partial v},$$

$$\frac{\partial f_{2k}}{\partial t} + 2 i v k f_{2k} = \frac{e}{m} \varepsilon_k \frac{\partial f_k}{\partial v} + \frac{e}{m} \varepsilon_{-k} \frac{\partial f_{3k}}{\partial v}, \text{ etc.}$$

They terminate this chain of equations by using the quasi-linear approximation [i.e., by neglecting the term $(e/m)\varepsilon_{-k}(\partial f_{2k}/\partial v)$ in the second equation]. However, we know from the breakdown of the linear theory that $(\partial f_k/\partial v) \sim (\partial f_0/\partial v)$ in the resonant region when $t \gtrsim \tau$. Extending this argument by iteration to the equation for f_{2k} shows that $(\partial f_{2k}/\partial v) \sim (\partial f_0/\partial v)$ in the resonant region when $t \gtrsim \tau$. Consequently, it is not apparent that the quasi-linear approximation is valid in the resonant region and that Altshul and Karpman have correctly solved the Vlasov equation to order $\gamma_L \tau$.

The main qualitative difference between Altshul and Karpman's results and the present results lies in the large t/τ behavior of $\gamma(t)$. Altshul and Karpman's $\gamma(t)$ oscillates indefinitely; whereas, the present $\gamma(t)$ phase mixes to zero as t/τ approaches infinity. The physical basis for the phase mixing will be presented in the next section.

III. PHYSICAL INTERPRETATION

The physical interpretation of the above results may easily be seen by considering the phase trajectories of the resonant electrons. In a coordinate system moving with the wave, the phase trajectories are as shown in Fig. 2.

To acquaint ourselves with Fig. 2, let us first consider the linear theory of collisionless damping. Only those electrons whose velocities are such that $kvt \lesssim 1$ can make any significant contribution to the rate of change of kinetic energy; so, we may confine our attention to those electrons between two tra-

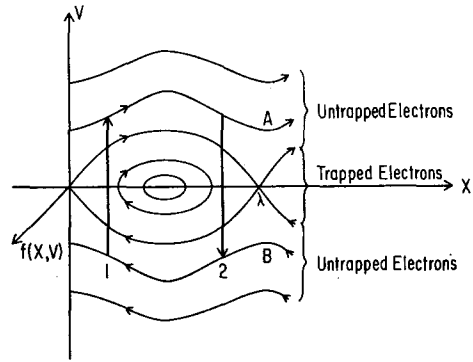


FIG. 2. The phase trajectories of the resonant electrons.

jectories such as A and B. Between these two trajectories, there is a net upward flow along vector 1 and a net downward flow along vector 2. For the case of damping, the initial distribution (plotted out of the paper in Fig. 2) decreases as a function of velocity. Consequently, a region of high density moves upward along vector 1 and a region of relatively low density moves downward along vector 2. This results in a net increase in the kinetic energy of the resonant electrons and a consequent damping of the wave.

To understand the nonlinear limit of collisionless damping, we should note that the first term on the right-hand side of Eq. (30) represents the untrapped electrons (see Fig. 2) and that the second term represents the trapped electrons. On a nonlinear time scale, the trapped electrons make complete cycles with a period of order τ . These electrons carry along the density from their original position and thus cause a cyclic variation in the density at any one point. It is this variation in density that causes the oscillatory behavior of the second term in Eq. (30). On a nonlinear time scale, the untrapped electrons are all "going over hill and dale" and, thus, produce the oscillatory behavior of the first term in Eq. (30).

The fact that $\gamma(t)$ phase mixes to zero as t/τ approaches infinity is easily explained in terms of an ergodic theorem. In Appendix B, a new distribution $\hat{f}$ is defined by coarse-grain-averaging the actual distribution. It is shown that as t/τ approaches infinity the coarse grain distribution becomes constant along the phase trajectories and the size of the mesh used in the coarse grain averaging goes to zero. Since the coarse grain distribution becomes constant along any phase trajectory, it also becomes a constant function of time. As the size of the mesh goes to zero, the integral of any smooth function weighted by the distribution will depend only on

³ L. M. Altshul and V. I. Karpman, presented at the Novosibirsk Conference on Many Body Theory (1965) (to be published).

the coarse grain distribution. Consequently, T becomes a constant and $\gamma(t)\alpha dT/dt$ approaches zero as t/τ approaches infinity.

Since the final coarse grain distribution is constant along any phase trajectory, its value on this trajectory can easily be obtained by averaging the initial distribution over this trajectory. In Appendix B, this procedure is carried out and the following integral is calculated:

$$\Delta T = \frac{nm}{2} \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{dx}{\lambda} \int_{-\infty}^{+\infty} dv \left(v + \frac{\omega}{k} \right)^2 (f - f_0).$$

By invoking conservation of energy, we can set ΔT equal to $-\Delta\mathcal{E}/8\pi = -\mathcal{E}_0^2/4\pi \int_0^\infty \gamma(t)dt$. In this fashion, we obtain a formula for $\int_0^\infty \gamma(t)dt$ that

$$\int_0^{\frac{1}{2}\pi} \frac{d\xi}{[1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} [1 - \rho^2 \sin^2(\xi)]^{\frac{1}{2}}} = \int_0^{\frac{1}{2}\pi} \frac{d\xi [\rho^4 - 2\kappa^2 \rho^2 - 2\rho^2 + 2\kappa^2 + 2\kappa^2 \rho^2 \sin^2(\xi) - \rho^4 \kappa^2 \sin^4(\xi)]}{2(1 - \rho^2)(\kappa^2 - \rho^2)[1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} [1 - \rho^2 \sin^2(\xi)]^{\frac{1}{2}}},$$

Eq. (29) can be written as

$$\begin{aligned} \frac{dT}{dt} = -\gamma_L \frac{\mathcal{E}^2}{4\pi} \left\{ \frac{64}{\pi} \int_0^1 \frac{d\kappa}{\kappa} \int_0^{\frac{1}{2}\pi} \frac{d\xi \operatorname{sn}(t/\kappa\tau, \kappa) \operatorname{cn}(t/\kappa\tau, \kappa) \sin^2(\xi) \cos^2(\xi)}{\pi [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} [1 - \kappa^2 \sin^2(t/\kappa\tau, \kappa) \sin^2(\xi)]} \right. \\ \left. + \frac{64}{\pi} \int_1^\infty \frac{d\kappa}{\kappa^5} \int_0^{\frac{1}{2}\pi} \frac{d\xi \operatorname{sn}(t/\tau, 1/\kappa) \operatorname{dn}(t/\tau, 1/\kappa) \sin^2(\xi) [1 - (1/\kappa^2) \sin^2(\xi)]^{\frac{1}{2}}}{\pi [1 - (1/\kappa^2) \sin^2(t/\tau, 1/\kappa) \sin^2(\xi)]} \right\}. \quad (35) \end{aligned}$$

The ξ and ζ integrations could now be performed, but it is more useful to Fourier analyze the time dependence of the integrands before these integrations are carried out. If we let $z \equiv t/\kappa\tau$, then the first integrand can be represented as

$$\begin{aligned} I(z) &\equiv \frac{\operatorname{sn}(z, \kappa) \operatorname{cn}(z, \kappa) \sin^2(\xi) \cos^2(\xi)}{[1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} [1 - \kappa^2 \sin^2(z, \kappa) \sin^2(\xi)]^{\frac{1}{2}}} \\ &= \sum_n b_n e^{in\pi z/2F}. \end{aligned}$$

The coefficients b_n are given by

$$b_n = \frac{1}{4F} \int_{-2F}^{2F} I(z) e^{-in\pi z/2F} dz. \quad (36)$$

Let C be the contour around the parallelogram whose corners are $(-2F, 2F, 2iF', 2iF' - 4F)$. By using the periodic properties of the Jacobi elliptic functions, we can write

$$\begin{aligned} \int_C I(z) e^{-in\pi z/2F} dz &= [1 + (-1)^n e^{\pi n F'/F}] \\ &\cdot \int_{-2F}^{2F} I(z) e^{-in\pi z/2F} dz. \quad (37) \end{aligned}$$

By using $\operatorname{sn}(iF' + u, \kappa) = 1/\kappa \operatorname{sn}(u, \kappa)$ and $\operatorname{sn}(iF' - 2F + u, \kappa) = -(1/\kappa) \operatorname{sn}(u, \kappa)$, it is easy to show that the roots of $[1 - \kappa^2 \sin^2(z, \kappa) \sin^2(\xi)]$ provide four

is identical to that given by Eq. (34). Thus, an argument based upon conservation of energy and the ergodic theorem gives an expression for $\int_0^\infty \gamma(t)dt$ that is identical to the expression obtained by integrating our generalized damping coefficient, $\gamma(t)$.

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APPENDIX A

With the aid of the following identity⁴

poles at $z = iF' \pm F(\kappa, \xi)$ and $z = iF' - 2F \pm F(\kappa, \xi)$. By using the double angle formula to evaluate the residues at these poles, we find

$$\begin{aligned} \int_C I(z) e^{-in\pi z/2F} dz &= -2\pi i \sin(\xi) \cos(\xi) \sin \left[\frac{\pi n F(\kappa, \xi)}{2F} \right] e^{\pi n F'/2F} [1 + (-1)^n] \\ &= \frac{-2\pi i \sin(\xi) \cos(\xi) \sin \left[\frac{\pi n F(\kappa, \xi)}{2F} \right] e^{\pi n F'/2F} [1 + (-1)^n]}{\kappa^2 [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}}}. \quad (38) \end{aligned}$$

By combining Eq. (36), (37), and (38), we find

$$I(z) = \sum_{n=0}^{\infty} \frac{2\pi \sin(\xi) \cos(\xi) \sin \left[\frac{\pi n F(\kappa, \xi)}{F} \right] e^{\pi n F'/F} \sin \left(\frac{\pi n z}{F} \right)}{F \kappa^2 [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} [1 + e^{2\pi n F'/F}]}.$$

If we put this expression for $I(z)$ into the first integral of Eq. (35), then this integral becomes

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 \frac{d\kappa 2\pi e^{\pi n F'/F} \sin \left[\frac{\pi n t}{\kappa F \tau} \right]}{\kappa^3 F [1 + e^{2\pi n F'/F}]} \\ \cdot \int_0^{\frac{1}{2}\pi} \frac{d\xi \sin(\xi) \cos(\xi) \sin \left[\frac{\pi n F(\kappa, \xi)}{F} \right]}{\pi [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}}}. \end{aligned}$$

⁴ W. Gröbner and N. Hofreiter, *Integraltafel* (Springer-Verlag, Berlin, 1963), Vol. 1, p. 63.

By using the substitution $u = F(\kappa, \xi)$, we can write the ξ -integration as

$$\int_0^{\frac{1}{2}\pi} \frac{d\xi \sin(\xi) \cos(\xi) \sin\left[\frac{\pi n F(\kappa, \xi)}{F}\right]}{\pi[1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}}} = \int_0^F \frac{du}{\pi} \operatorname{sn}(u, \kappa) \operatorname{cn}(u, \kappa) \sin\left[\frac{\pi nu}{F}\right].$$

The integral on the right-hand side is an even function of u and can be written as

$$\int_0^F \frac{du}{\pi} \operatorname{sn}(u, \kappa) \operatorname{cn}(u, \kappa) \sin\left[\frac{\pi nu}{F}\right] = \frac{1}{2i} \int_{-F}^F \frac{du}{\pi} \operatorname{sn}(u, \kappa) \operatorname{cn}(u, \kappa) e^{i\pi nu/F}.$$

If we let C' be the contour around the rectangle whose corners are $(-F, F, F + 2iF')$, and $(-F + 2iF')$, then the periodic properties of the Jacobi elliptic functions allow us to write

$$\int_{C'} du \operatorname{sn}(u, \kappa) \operatorname{cn}(u, \kappa) e^{i\pi nu/F} = [1 + e^{-\pi 2nF'/F}] \int_{-F}^F du \operatorname{sn}(u, \kappa) \operatorname{cn}(u, \kappa) e^{i\pi nu/F}.$$

Picking up the pole at $u = iF'$ gives

$$\int_0^{\frac{1}{2}\pi} \frac{d\xi \sin(\xi) \cos(\xi) \sin\left[\frac{\pi n F(\kappa, \xi)}{F}\right]}{\pi[1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}}} = \frac{\pi n e^{-\pi n F'/F}}{\kappa^2 F [1 + e^{-2\pi n F'/F}]}.$$

Consequently, the first integral in Eq. (35) can be written as

$$\sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 \frac{d\kappa}{\kappa^5} \frac{2n\pi^2 \sin\left[\frac{\pi n t}{\kappa F \tau}\right]}{F^2(1 + q^{2n})(1 + q^{-2n})}, \quad (39)$$

where

$$q \equiv \exp\left[\frac{\pi F'}{F}\right].$$

The second integral in Eq. (35) can be transformed in a very similar fashion. If we also replace $1/\kappa$ by κ in the second integral, then this integral becomes

$$\sum_{n=0}^{\infty} \frac{64}{\pi} \int_0^1 d\kappa \frac{(2n+1)\pi^2 \kappa \sin\left[\frac{(2n+1)\pi t}{2F\tau}\right]}{F^2(1 + q^{2n+1})(1 + q^{-2n-1})}. \quad (40)$$

Combining Eqs. (35), (39), and (40) gives the result quoted in Eq. (30).

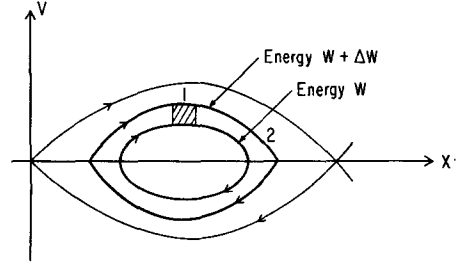


FIG. 3. Initial distribution used in the proof of ergodic theorem.

APPENDIX B

To prove the ergodic theorem of Appendix A, consider an initial distribution limited to the small patch of phase space represented by the shaded area in Fig. 3. The electrons with energy W have a shorter period than those with energy $W + \Delta W$. Eventually, the electrons with energy W "lap" the electrons with energy $W + \Delta W$, and the shaded area is stretched into a long, fibrous spiral that fills the region between the two trajectories, W and $W + \Delta W$. Let $\hat{f}$ be a coarse grain distribution obtained by averaging the actual distribution over phase elements which are large enough to be traversed by many strands of the spiral.

Conservation of phase density demands that in the time-asymptotic limit the rate of flow of phase density past point 1 (see Fig. 3) equals the rate of flow past point 2. The rate of flow past point 1 is $\hat{f}(1)v_1\Delta v_1$, where $\Delta W = \partial W/\partial v_1\Delta v_1$. The rate of flow past point 2 is

$$\hat{f}(2)[v_2^2 + v_2^2]^{\frac{1}{2}}[(\Delta x_2)^2 + (\Delta v_2)^2]^{\frac{1}{2}},$$

where $(v_2, \dot{v}_2)$ is a vector perpendicular to $(\Delta x_2, \Delta v_2)$ and $\Delta W = \partial W/\partial x_2\Delta x_2 + \partial W/\partial v_2\Delta v_2$. Equating the two flow rates gives

$$\hat{f}(1)v_1\Delta v_1 = \hat{f}(2)[v_2^2 + v_2^2]^{\frac{1}{2}}[(\Delta x_2)^2 + (\Delta v_2)^2]^{\frac{1}{2}}. \quad (41)$$

Using $\partial W/\partial v = mv$ and $\partial W/\partial x = -m\dot{v}$ and equating the two expressions for ΔW gives

$$v_1\Delta v_1 = -\dot{v}_2\Delta x_2 + v_2\Delta v_2. \quad (42)$$

Since the vectors $(v_2, \dot{v}_2)$ and $(\Delta x_2, \Delta v_2)$ are perpendicular, we can equate

$$-\dot{v}_2\Delta x_2 + v_2\Delta v_2 = [v_2^2 + v_2^2]^{\frac{1}{2}}[(\Delta x_2)^2 + (\Delta v_2)^2]^{\frac{1}{2}}. \quad (43)$$

Combining Eqs. (41), (42), and (43) immediately gives $\hat{f}(1) = \hat{f}(2)$. Since point 2 was any arbitrary point along the trajectory, we have shown that the coarse-grain distribution is constant along the phase trajectory. From the definition of the coarse-grain distribution, it also follows that the size of the mesh used in the coarse-grain-averaging process goes to zero as t/τ approaches infinity.

For a periodic initial distribution, one can extend the above arguments to the untrapped electrons and show that the coarse grain distribution becomes a constant along any trajectory.

To find the value of the coarse-grain distribution on any trajectory, we must average the initial distribution over this trajectory. In the resonant region, the initial waveframe distribution can be written as

$$f_0(v) = f_0(0) + \partial f_0 / \partial v |_{v=0} v + \dots \quad (44)$$

Specifying the trajectories by κ rather than W gives for $\kappa^2 < 1$

$$\begin{aligned} \bar{f} &\equiv \frac{\Delta N}{\Delta A} = \int_0^\lambda f_0(v) \Delta v(x) dx \left[\int_0^\lambda \Delta v(x) dx \right]^{-1} \\ &= f_0(0) + \frac{\partial f_0}{\partial v} \frac{\pi}{F k \kappa \tau}, \end{aligned} \quad (45)$$

where $\Delta W = (\partial W / \partial v) \Delta v(x)$ and κ takes the sign of v . For $\kappa^2 > 1$ one obviously finds $\bar{f} = f_0(0)$. The difference between the initial and final distributions is given by

$$\begin{aligned} \bar{f} - f_0 &= (\partial f_0 / \partial v) (2 / \kappa k \tau) \{ (\pi / 2F) - [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} \}, \\ &\quad \text{for } \kappa^2 < 1, \\ &= (\partial f_0 / \partial v) (2 / \kappa k \tau) \{ - [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} \}, \\ &\quad \text{for } \kappa^2 > 1. \end{aligned} \quad (46)$$

The change in kinetic energy of the resonant electrons is

$$\begin{aligned} \Delta T &= \frac{nm}{2} \int_{-\frac{1}{2}\lambda}^{\frac{1}{2}\lambda} \frac{dx}{\lambda} \int_{-\infty}^{+\infty} dv \left(v + \frac{\omega}{k} \right)^2 (\bar{f} - f_0) \\ &= \frac{16nm\omega}{k^3} \int_0^{\frac{1}{2}\pi} \frac{d\xi}{\pi} \int_0^\infty d\xi \xi (\bar{f} - f_0). \end{aligned} \quad (47)$$

Changing coordinates from (ξ, ξ) to (ξ, κ) gives

$$\begin{aligned} \Delta T &= \frac{16nm\omega}{k^3} \left\{ \int_0^1 \frac{d\kappa}{\kappa^{\frac{3}{2}} \tau^2} \int_0^{\frac{1}{2}\pi} \frac{d\xi}{\pi} (\bar{f} - f_0) \right. \\ &\quad \left. + \int_1^\infty \frac{d\kappa}{\kappa^{\frac{3}{2}} \tau^2} \int_0^{\sin^{-1}(1/\kappa)} \frac{d\xi}{\pi} (\bar{f} - f_0) \right\}. \end{aligned} \quad (48)$$

Substituting Eq. (46) into Eq. (48) and using the transformation given in Eq. (17) gives

$$\begin{aligned} \Delta T &= \frac{32nm\omega}{\pi \tau^3 k^4} \frac{\partial f_0}{\partial v} \\ &\quad \cdot \left\{ \int_0^1 \frac{d\kappa}{\kappa^4} \int_0^{\frac{1}{2}\pi} d\xi \left[\frac{\pi}{2F} - [1 - \kappa^2 \sin^2(\xi)]^{\frac{1}{2}} \right] \right. \\ &\quad \left. - \int_1^\infty \frac{d\kappa}{\kappa^5} \int_0^{\frac{1}{2}\pi} d\xi \frac{\cos^2(\xi)}{[1 - (1/\kappa^2) \sin^2(\xi)]^{\frac{1}{2}}} \right\}. \end{aligned} \quad (49)$$

By carrying out the ξ and ζ integrations and by replacing $1/\kappa$ by κ in the second integral, we find

$$\begin{aligned} \Delta T &= -\gamma_L \tau \frac{\varepsilon^2}{4\pi} \frac{64}{\pi} \\ &\quad \cdot \int_0^1 d\kappa \left\{ \frac{1}{\kappa^4} \left[\frac{E}{\pi} - \frac{\pi}{4F} \right] + \frac{\kappa}{\pi} [E + (\kappa^2 - 1)F] \right\}, \end{aligned}$$

where $E \equiv E(\kappa, \frac{1}{2}\pi)$ is the complete elliptic integral of the second kind. Invoking conservation of energy gives

$$\begin{aligned} \int_0^\infty \gamma(t) dt &= \gamma_L \tau \frac{64}{\pi} \\ &\quad \cdot \int_0^1 d\kappa \left\{ \frac{1}{\kappa^4} \left[\frac{E}{\pi} - \frac{\pi}{4F} \right] + \frac{\kappa}{\pi} [E + (\kappa^2 - 1)F] \right\}. \end{aligned} \quad (50)$$

To compare the above results with Eq. (34), we must first sum the series in Eq. (34). The Fourier series representation of $\text{cn}(u, \kappa)$ is

$$\text{cn}(u, \kappa) = \frac{2\pi}{F\kappa} \sum_{n=0}^\infty \frac{q^{n+\frac{1}{2}}}{1 + q^{2n+1}} \cos \left[\frac{(2n+1)\pi u}{2F} \right]. \quad (51)$$

Multiplying both sides of Eq. (51) by $\text{cn}(u, \kappa)$ and integrating from $-F$ to F gives

$$\frac{\kappa}{\pi} [E + (\kappa^2 - 1)F] = \frac{2\pi\kappa}{F} \sum_{n=0}^\infty \frac{1}{(1 + q^{2n+1})(1 + q^{-2n-1})}. \quad (52)$$

Repeating the same procedure with $\text{dn}(u, \kappa)$ instead of $\text{cn}(u, \kappa)$ gives

$$\left[\frac{E}{\pi} - \frac{\pi}{4F} \right] = \frac{2\pi}{F} \sum_{n=1}^\infty \frac{1}{(1 + q^{2n})(1 + q^{-2n})}. \quad (53)$$

Substituting Eq. (52) and (53) into Eq. (50) reduces Eq. (50) to the result given in Eq. (34).